

\$IfNot ParametricTable

i=2300

\$EndIf

T_Evap_In = **lookup**('FromExcel', i, '21:t_gshp_evap_out')
 T_Evap_Out = **lookup**('FromExcel', i, '20:t_gshp_evap_in')
 T_Cond_In = **lookup**('FromExcel', i, '15:t_gshp_compsr_in')
 T_Cond_Out = **lookup**('FromExcel', i, '13:t_gshp_hex_in_r32')
 Work_All = **lookup**('FromExcel', i, '4:pow_gshp_gshp_ac')
 R\$ = 'R32'
 x[3] = 0
 P_High_Stage = 3301.325
 P_Low_Stage = 941.325

COP = 3.9

T[1] = T_Evap_Out
 P[1] = P[4]
 h[1] = **enthalpy**(R\$, T=T[1], P=P[1])
 s[1] = **entropy**(R\$, T=T[1], P=P[1])

T[2] = T_Cond_In
 s[2] = s[1]
 P[2] = P[3]
 h2_Maximum = **enthalpy**(R\$, T=T[2], P=P[2])
 h2_Minimum = h[1]
 P_max = **pressure**(R\$, T=T[2], h=h[1])
 h[2] = h2_Maximum
 T[3] = T_Cond_Out
 P[3] = **p_sat**(R\$, T=T[3])

h[3]=h[4]
 T[4] = T_Evap_In
 P[4] = **p_sat**(R\$, T=T[4])
 COP = (h[1]-h[4])/(h[2] - h[1])

m_dot*h[1] = m_dot*h[2]-Work_All
 Q_Evap + m_dot*h[4] = m_dot*h[1]
 Q_Cond +m_dot*h[2] = m_dot*h[3]

SOLUTION

Unit Settings: SI C kPa kJ mass deg

$$\text{COP} = 3.9$$

$$h_{2\text{Minimum}} = 522.3$$

$$\dot{m} = 0.1413 \text{ [kg/s]}$$

$$P_{\text{Low,Stage}} = 941.3$$

$$Q_{\text{Cond}} = -4.361 \text{ [kW]}$$

$$R\$ = \text{'R32'}$$

$$T_{\text{Cond,Out}} = 29.61$$

$$T_{\text{Evap,Out}} = 12.4 \text{ [C]}$$

$$h_{2\text{Maximum}} = 528.6$$

$$i = 2300$$

$$P_{\text{High,Stage}} = 3301$$

$$P_{\text{max}} = 2104$$

$$Q_{\text{Evap}} = 3.471 \text{ [kW]}$$

$$T_{\text{Cond,In}} = 37.72$$

$$T_{\text{Evap,In}} = 7.9 \text{ [C]}$$

$$\text{Work}_{\text{All}} = 0.89 \text{ [kW]}$$

11 potential unit problems were detected.

Arrays Table: Main

	x_i	T_i [C]	P_i [kPa]	h_i [kJ/(kg*K)]	s_i
1		12.4	1040	522.3	2.147
2		37.72	1908	528.6	2.147
3	0	29.61	1908	497.7	
4		7.9	1040	497.7	



